Initial Data Visualization

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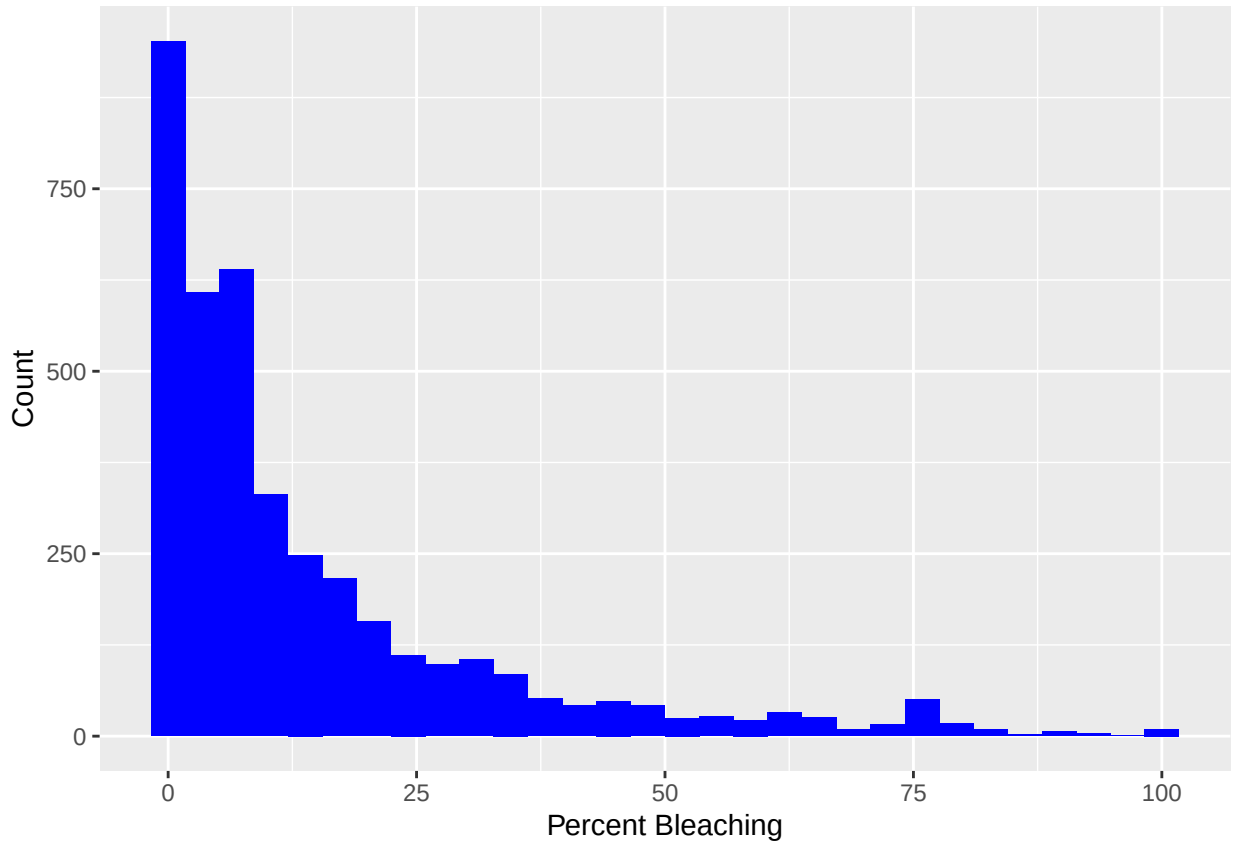
Setup

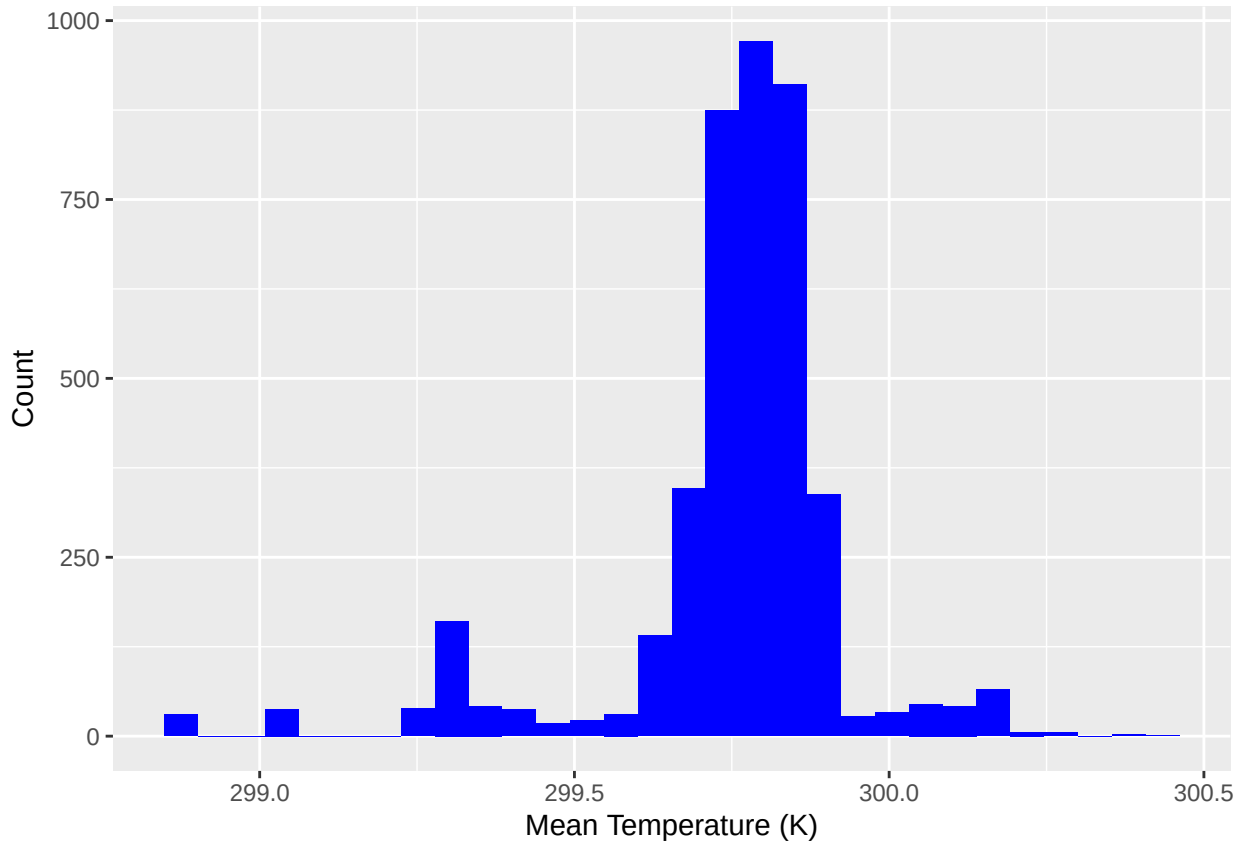
```r
# Load necessary packages
library(here)
library(tidyverse)
library(ggplot2)
library(readr)
library(scrutiny)
library(patchwork)


# Load dataset
bleaching <- read.csv("Code/bahamas_florida_bleaching_subset.csv")
bleaching$Percent_Bleaching <- as.numeric(parse_number(bleaching$Percent_Bleaching))
bleaching$Temperature_Mean <- as.numeric(parse_number(bleaching$Temperature_Mean))
bleaching$Turbidity <- as.numeric(parse_number(bleaching$Turbidity))
bleaching$Distance_to_Shore <- as.numeric(parse_number(bleaching$Distance_to_Shore))
```

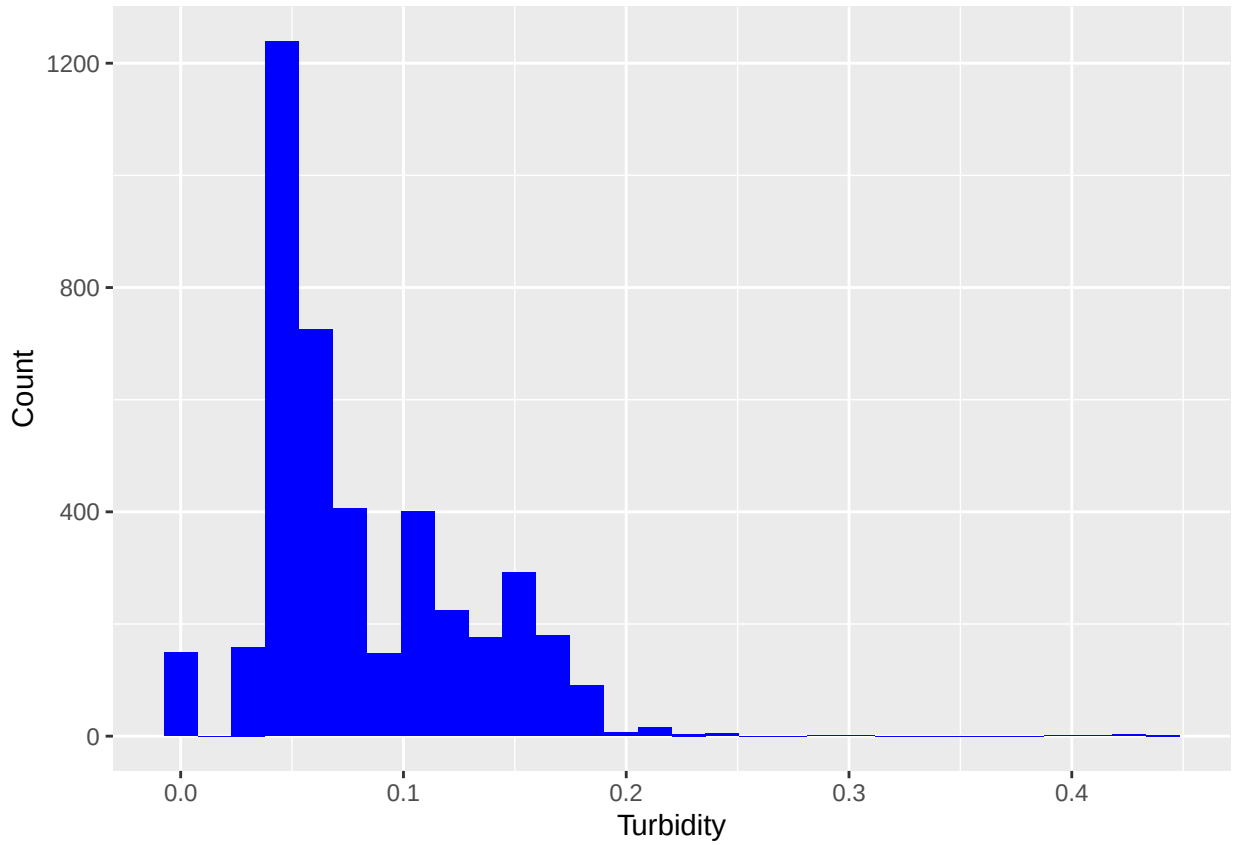
```r
# Histograms
hist_bleaching <- ggplot(bleaching, aes(x = Percent_Bleaching)) +
  geom_histogram(fill = "blue") +
  labs(x = "Percent Bleaching",
       y = "Count")
hist_bleaching
```



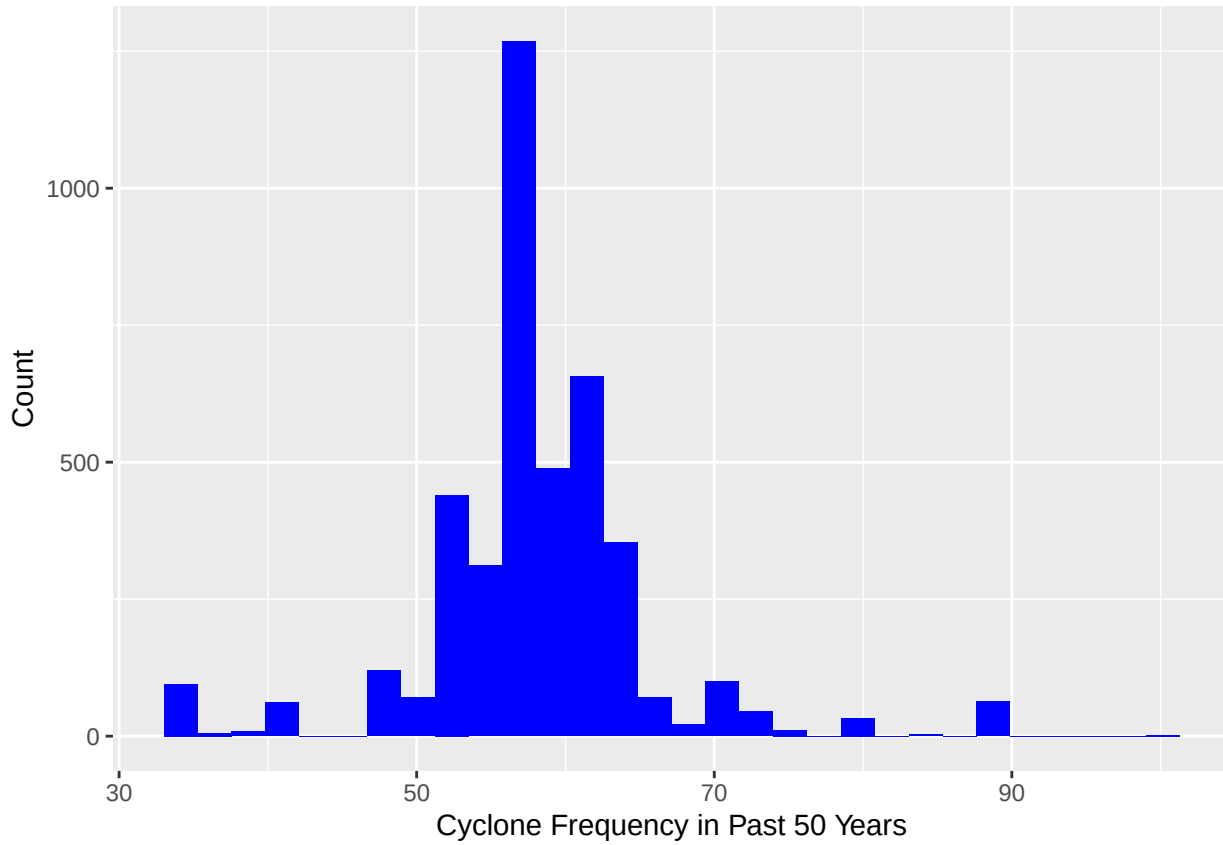

```
hist_temp <- ggplot(bleaching, aes(x = Temperature_Mean)) +
  geom_histogram(fill = "blue") +
  labs(x = "Mean Temperature (K)",
       y = "Count")
hist_temp
```



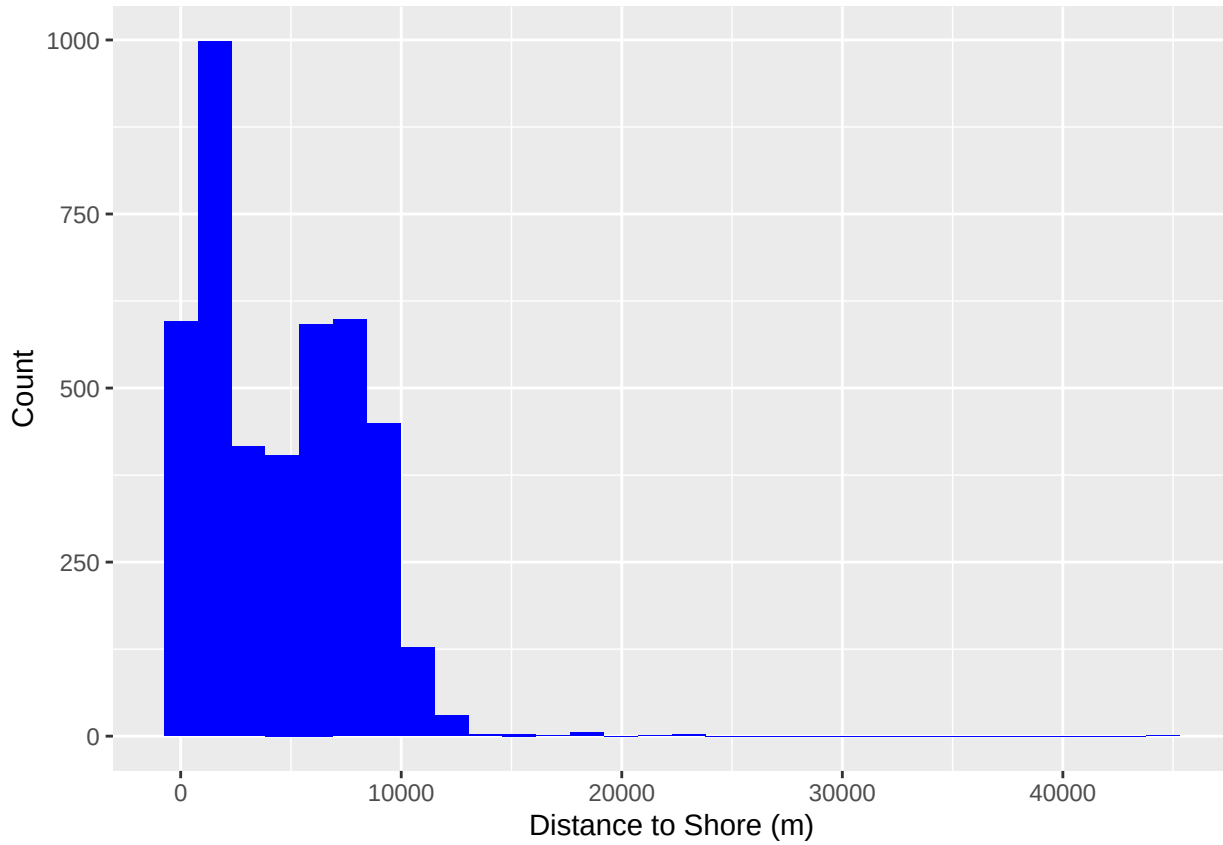

```
hist_turbidity <- ggplot(bleaching, aes(x = Turbidity)) +
  geom_histogram(fill = "blue") +
  labs(y = "Count")
hist_turbidity
```



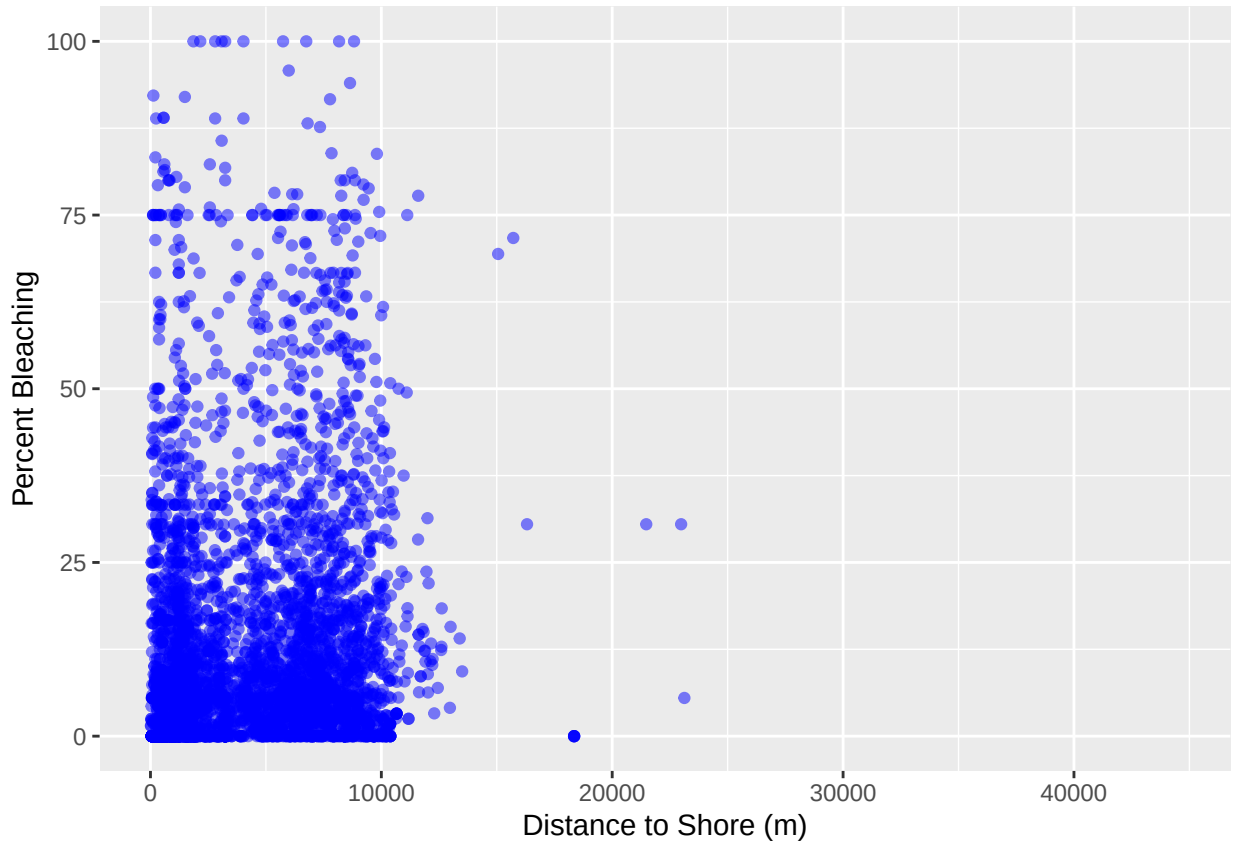

```
hist_cyclone <- ggplot(bleaching, aes(x = Cyclone_Frequency)) +
  geom_histogram(fill = "blue")+
  labs(x = "Cyclone Frequency in Past 50 Years",
       y = "Count")
hist_cyclone
```



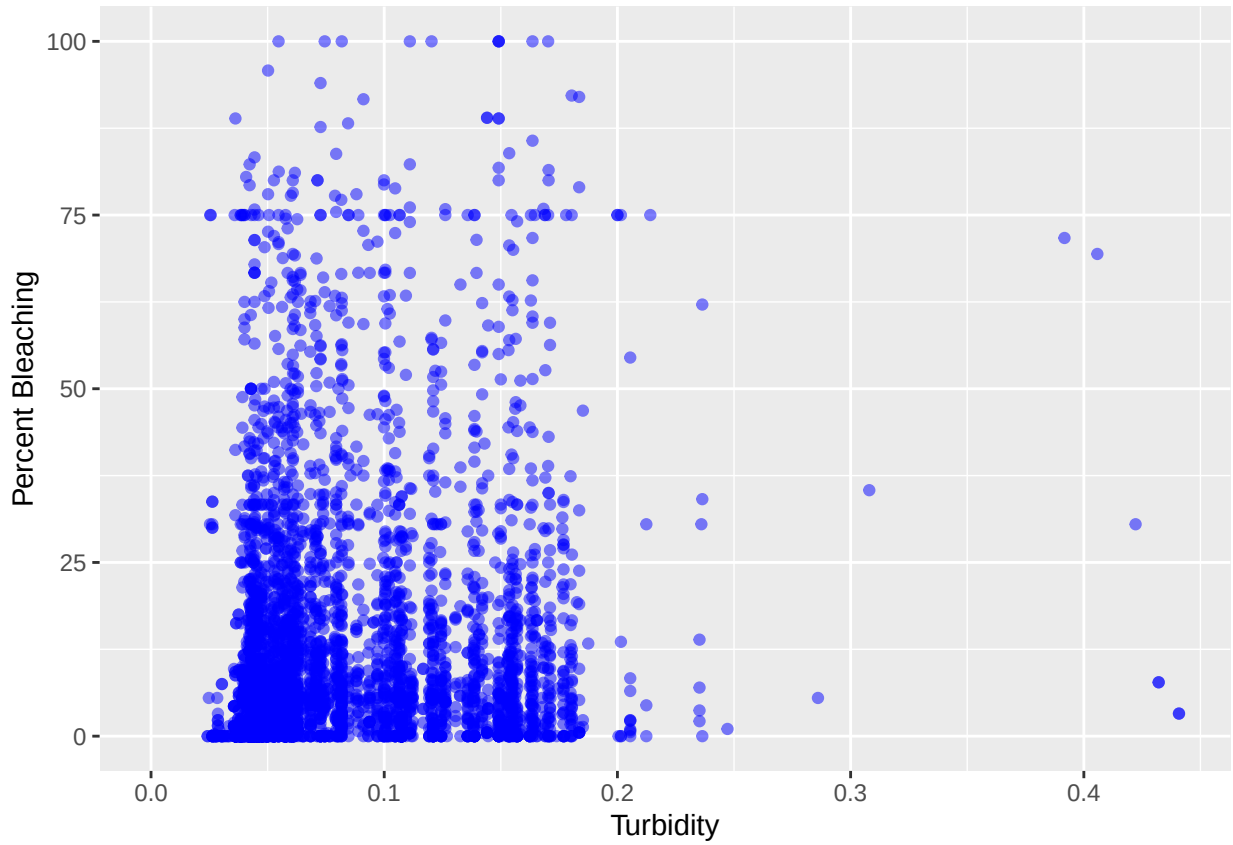

```
hist_dist_to_shore <- ggplot(bleaching, aes(x = Distance_to_Shore)) +
  geom_histogram(fill = "blue") +
  labs(x = "Distance to Shore (m)",
       y = "Count")
hist_dist_to_shore
```



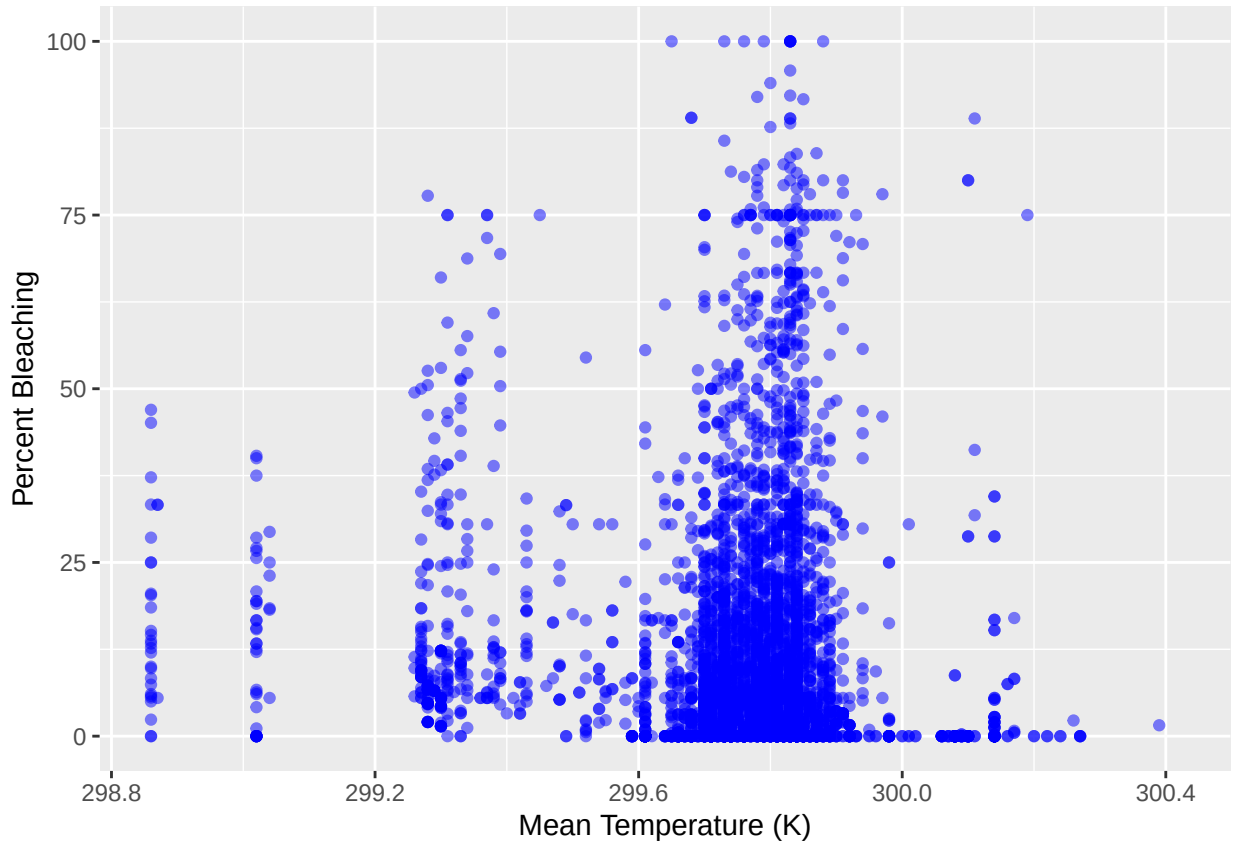

```r
# Scatter plots
scatter_distance_to_shore <- ggplot(bleaching, aes(x = Distance_to_Shore,
                                                   y = Percent_Bleaching)) +
  geom_point(color = "blue", alpha = 0.5) +
  labs(x = "Distance to Shore (m)",
       y = "Percent Bleaching")
scatter_distance_to_shore
```



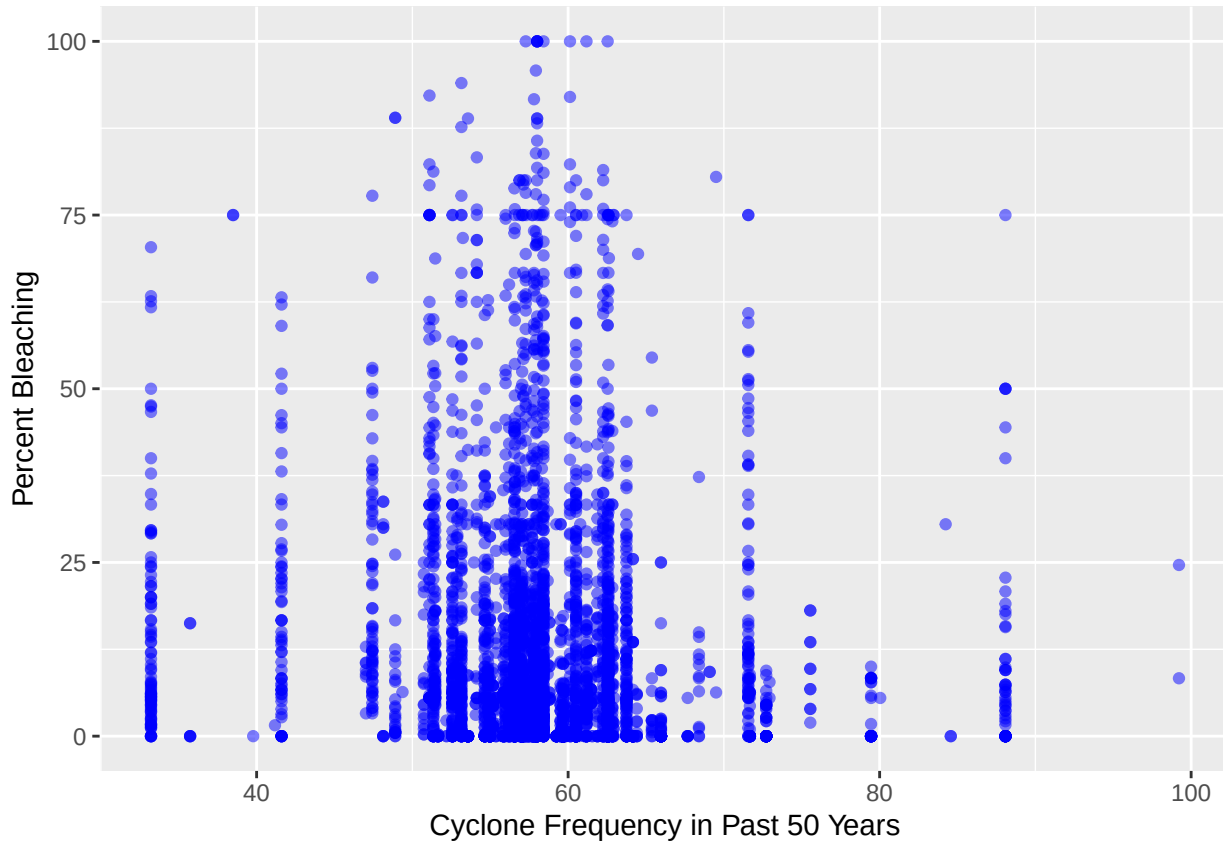

```
scatter_turbidity <- ggplot(bleaching, aes(x = Turbidity,
                                           y = Percent_Bleaching)) +
  geom_point(color = "blue", alpha = 0.5) +
  labs(y = "Percent Bleaching")
scatter_turbidity
```




```
scatter_temp <- ggplot(bleaching, aes(x = Temperature_Mean,
                                      y = Percent_Bleaching)) +
  geom_point(color = "blue", alpha = 0.5) +
  labs(x = "Mean Temperature (K)",
       y = "Percent Bleaching")
scatter_temp
```




```
scatter_cyclone <- ggplot(bleaching, aes(x = Cyclone_Frequency,
                                         y = Percent_Bleaching)) +
  geom_point(color = "blue", alpha = 0.5) +
  labs(x = "Cyclone Frequency in Past 50 Years",
       y = "Percent Bleaching")
scatter_cyclone
```




```r
# Multi-collinearity
cor.test(bleaching$Percent_Bleaching, bleaching$Temperature_Mean) # not correlated
```

```
##
##  Pearson's product-moment correlation
##
## data:  bleaching$Percent_Bleaching and bleaching$Temperature_Mean
## t = -1.3485, df = 3986, p-value = 0.1776
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.052356921  0.009690898
## sample estimates:
##         cor
## -0.02135357
```

```r
cor.test(bleaching$Percent_Bleaching, bleaching$Turbidity) # not correlated
```

```
##
##  Pearson's product-moment correlation
##
## data:  bleaching$Percent_Bleaching and bleaching$Turbidity
## t = 5.8193, df = 3987, p-value = 6.376e-09
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.06091086 0.12245693
```

```
## sample estimates:
##        cor
## 0.09177154
```

```r
cor.test(bleaching$Percent_Bleaching, bleaching$Cyclone_Frequency) # not correlated
```

```
##
##  Pearson's product-moment correlation
##
## data:  bleaching$Percent_Bleaching and bleaching$Cyclone_Frequency
## t = -5.4171, df = 3988, p-value = 6.415e-08
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.11618889 -0.05458125
## sample estimates:
##         cor
## -0.08546676
```

```r
cor.test(bleaching$Percent_Bleaching, bleaching$Distance_to_Shore) # not correlated
```

```
##
##  Pearson's product-moment correlation
##
## data:  bleaching$Percent_Bleaching and bleaching$Distance_to_Shore
## t = 6.8093, df = 3986, p-value = 1.127e-11
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.07644849 0.13781147
## sample estimates:
##       cor
## 0.1072321
```

```r
cor.test(bleaching$Temperature_Mean, bleaching$Turbidity) # not correlated
```

```
##
##  Pearson's product-moment correlation
##
## data:  bleaching$Temperature_Mean and bleaching$Turbidity
## t = -4.5204, df = 4223, p-value = 6.342e-06
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.09934009 -0.03932052
## sample estimates:
##        cor
## -0.0693931
```

```r
cor.test(bleaching$Temperature_Mean, bleaching$Cyclone_Frequency) # not correlated
```

```
##
##  Pearson's product-moment correlation
##
```

```
## data:  bleaching$Temperature_Mean and bleaching$Cyclone_Frequency
## t = -0.27902, df = 4223, p-value = 0.7802
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.03444398  0.02586463
## sample estimates:
##          cor
## -0.004293578
```

```r
cor.test(bleaching$Temperature_Mean, bleaching$Distance_to_Shore) # not correlated
```

```
##
##  Pearson's product-moment correlation
##
## data:  bleaching$Temperature_Mean and bleaching$Distance_to_Shore
## t = 15.4, df = 4221, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.2018893 0.2590070
## sample estimates:
##       cor
## 0.2306468
```

```r
cor.test(bleaching$Turbidity, bleaching$Cyclone_Frequency) # not correlated
```

```
##
##  Pearson's product-moment correlation
##
## data:  bleaching$Turbidity and bleaching$Cyclone_Frequency
## t = -1.3658, df = 4224, p-value = 0.1721
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.051129255  0.009146729
## sample estimates:
##         cor
## -0.02101035
```

```r
cor.test(bleaching$Turbidity, bleaching$Distance_to_Shore) # not correlated
```

```
##
##  Pearson's product-moment correlation
##
## data:  bleaching$Turbidity and bleaching$Distance_to_Shore
## t = 8.4782, df = 4222, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.09961356 0.15892161
## sample estimates:
##       cor
## 0.1293833
```

```r
cor.test(bleaching$Cyclone_Frequency, bleaching$Distance_to_Shore) # not correlated
```

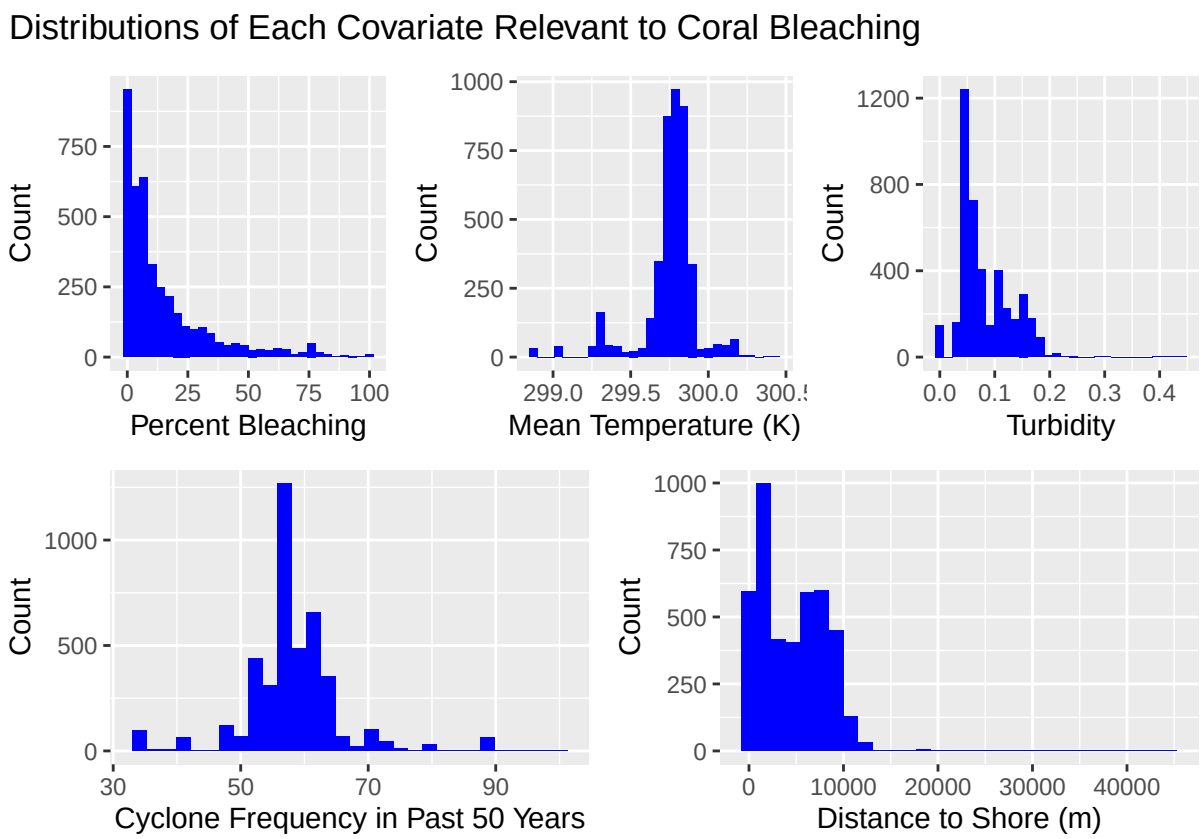
```
##
##  Pearson's product-moment correlation
##
## data:  bleaching$Cyclone_Frequency and bleaching$Distance_to_Shore
## t = -2.6928, df = 4223, p-value = 0.007113
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.07146784 -0.01126141
## sample estimates:
##         cor
## -0.04140221
```

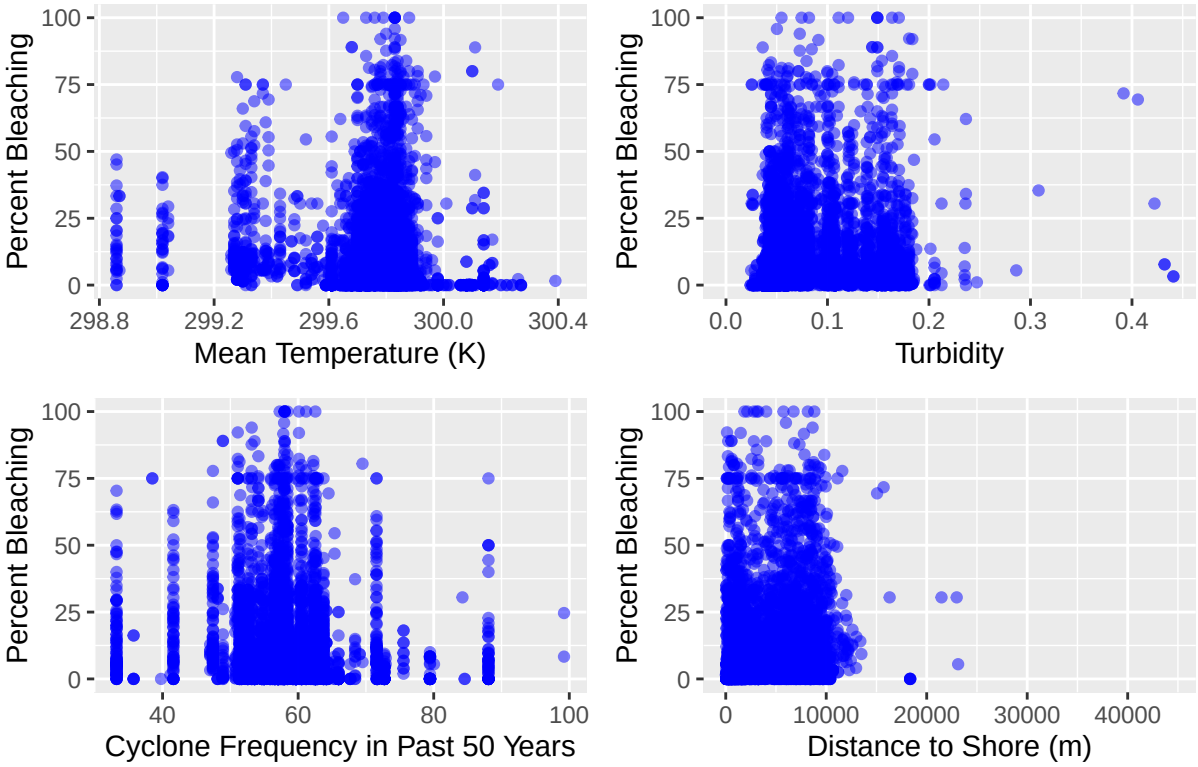
Notes from Clay

percent data: look into GLM family that people use. probably not binomial. gaussian?

Add spatial component. could be used to add categorical variable

```r
# Histograms
histograms <- (hist_bleaching + hist_temp + hist_turbidity) / (hist_cyclone + hist_dist_to_shore) +
  plot_annotation(
  title = "Distributions of Each Covariate Relevant to Coral Bleaching")
histograms
```




```r
# Scatters
scatterplots <- (scatter_temp + scatter_turbidity) / (scatter_cyclone + scatter_distance_to_shore) +
  plot_annotation(
    title = "Relationship Between Covariates and Percent Bleaching")
scatterplots
```

Relationship Between Covariates and Percent Bleaching